TT-S-222b
November 17, 1967
SUPERSEDING
Fed. Spec. TT-S-222a
July 22, 1965

FEDERAL SPECIFICATION

SEALING COMPOUND: ASBESTOS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of asbestos sealing compound for deep water fording kits and for protecting certain truck engines against water (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-704 - Pails: Shipping, Steel (1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8030

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM) Standards:

D 92 - Flash and Fire Points by Cleveland Open Cup.
D 95 - Water in Petroleum and other Bituminous Materials.
D 128 - Analysis of Lubricating Greases.
D 217 - Cone Penetration of Lubricating Grease.
D 270 - Sampling Petroleum and Petroleum Products.
D 445 - Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities).
D 566 - Dropping Point of Lubricating Grease.
D 942 - Oxidation Stability of Lubricating Greases by the Oxygen Bomb Method.
D 2161 - Conversion of Kinematic Viscosity to Saybolt Universal Viscosity or to Saybolt Furol Viscosity.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Materials. The asbestos sealing compound shall consist of a mineral oil, thickened with a calcium soap of one or more higher fatty acids, to which has been added asbestos floats.

3.2 Physical requirements of mineral oil. The physical requirements of the mineral oil extracted from the sealing compound shall be as specified in table I when the oil is tested as specified in 4.5.1.

TABLE I. Requirements of mineral oil

Property	Requirement
Viscosity at 100°F.(37.78°C.)	
Centistokes:	32 - 36.5
S.U.S. 1/:	155 - 175
Flash point, minimum	350°F.(176.7°C.)

1/ (Saybolt Universal Viscosity) For information only.

3.3 Physical requirements of asbestos sealing compound. The physical requirements of the asbestos sealing compound shall be as specified in table II and subsequent paragraphs when tested as specified in 4.5.2.

TABLE II. Requirements of asbestos sealing compound

Property	Requirement
Mineral Oil, percent, minimum:	51.5
Calcium soap, percent, maximum:	25
Water, percent, maximum:	2.5
Asbestos content, percent:	19 - 21
Worked penetration, 60 strokes:	110 - 150
Dropping point, minimum:	250°F. (121°C.)
Oxidation stability, p.s.i. pressure drop in 100 hours, maximum:	10

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3.4 Asbestos characteristics. The average length of the asbestos fibers used in the asbestos sealing compound shall be 0.0625 (1/16) inches. Less than 25 percent of the fibers shall be shorter than 0.0156 (1/64) inches, and less than 25 percent of the fibers shall be longer than 0.250 (1/4) inches in length.

3.5 Odor. The asbestos sealing compound shall have only a slight odor of mineral oil when sniffed. Any odor of rancidity or perfume shall be cause for rejection.

3.6 Workmanship. The asbestos sealing compound shall be free from lumps and shall be uniform in appearance when examined visually.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all asbestos sealing compound made from the same batch or blending operation, from the same raw materials, by one manufacturer, and offered for delivery at one time.

4.3 Sampling.

4.3.1 Sampling for examination of the preparation for delivery. Unless otherwise specified, a random sample of containers fully prepared for delivery shall be taken from each lot in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with the requirements of section 5 of this specification.

4.3.2 Sampling for test purposes. Unless otherwise specified in the contract or order, sampling of a lot for test purposes shall be made in accordance with ASTM D 270.

4.4 Examination.

4.4.1 Visual. Samples selected in accordance with 4.3.1 shall be examined for defects at the acceptable quality levels (AQLs) shown in table III. Any container in the sample having one or more defects or under the required fill, shall be rejected and if the number of rejected containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.2 Packaging, packing, and marking. Examination for packaging, packing, and marking shall be for the defects and at the AQLs shown in table III.

4.5.1 Extracted mineral oil. The mineral oil extracted from the asbestos sealing compound as specified in 4.5.2, shall be tested in accordance with the test methods listed in table IV.

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TABLE III. Method of examination

Category	Material	AQL percent	Defect	Method of inspection
Critical	None defined			
Major 101	Sealing compound: asbestos (see 3.5, 3.6, and 4.4.1)	2.5	Odor of rancidity or perfume	Olfactory
Major 102			Not free from lumps	Visual
Major 103			Not uniform in appearance	Visual
Major 104	Containers open (see 4.4.2, 5.1.1, or 5.1.2)		Improper size	Visual
Major 105		2.5	Improper type	Visual
Major 106			Improper fill 1/	Visual and scale 2/
Major 107	Container closed (see 4.4.2 5.1.1, 5.1.2, and 5.3)	2.5	Improperly closed	Visual
Major 108			Marking misleading or unidentifiable	Visual

1/ A container filled with the minimum required quantity of material shall be weighted, and this weight compared with the weight of unopened containers to determine fill.

2/ The scale used shall be approved by the procuring activity.

TABLE IV. Test methods, extracted mineral oil

Test	ASTM designation
Viscosity at 100°F. (37.78°C.)	
Centistokes:	D 445
Conversion of Kinematic to Saybolt viscosity:	D 2161
Flash point:	D 92

4.5.2 Asbestos sealing compound. Samples of asbestos sealing compound selected in accordance with 4.3.2 shall be tested in accordance with the test methods listed in table V and in 4.5.3.

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TABLE V. Test method, asbestos sealing compound

Test	ASTM designation
Mineral oil, percent:	D 128
Calcium soap, percent:	D 128
Water, percent:	D 95
Asbestos content:	D 128
Worked penetration:1/	D 217
Dropping	D 566
Oxidation stability:	D 942

1/ A mechanically operated grease worker shall be used to work the asbestos sealing compound for 60 strokes.

4.5.3 Asbestos characteristics. A representative portion of the asbestos fibers removed from the asbestos sealing compund as specified in 4.5.2, shall be examined under a low-power microscope equipped with a Filar eyepiece for conformance with the requirements in 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), the asbestos sealing compound shall be packaged 35 pounds in a pail conforming to PPP-P-704, type II, class 3.

5.1.2 Level C. Asbestos sealing compound shall be packaged in accordance with suppliers' commercial practice. Protection shall be such as to prevent deterioration during shipment and ensure safe delivery at destination.

5.2 Packing. Unless otherwise specified (see 6.2), material packaged as specified in 5.1 will require no further packing.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The asbestos sealing compound covered by this specification is intended for use as a sealing compound with deep water fording kits to cover sealing tape when more than a brush or spray coating is desired. It must not be used where temperatures above 225°F. (107°C.) will be encountered. This sealing compound may also be used to fill recesses around spark plugs on certain truck engines to prevent accumulation of water during storage.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size and type container desired if other than 35 pound pails.
(c) Applicable level of packaging required (see 5.1).
(d) When packing other than specified is required (see 5.2).

MILITARY CUSTODIANS:

Army - MR

Preparing activity:

Army - MR

Review activities:

MR, WC, AT, IS, MD

User activities:

YD, ME

Orders for the publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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